

Software Requirements Specification (SRS)

Selenium-Tested Performance Review Portal

Student: Jawad Shaikh | Roll No: 23102B0009 | Semester VII | Department of Computer Engineering

Table of Contents

1. Introduction
2. Problem Statement
3. Objectives
4. Scope
5. Stakeholders
6. Agile Planning
7. Product Backlog
8. User Stories
9. Sprint Plan
10. Kanban
11. Gantt
12. DevOps Lifecycle
13. Functional Requirements
14. Non Functional Requirements
15. Use Case
16. ER Diagram
17. Class Diagram
18. Architecture
19. API Design
20. Database
21. Risk Analysis
22. Future Scope
23. Conclusion

1. Introduction

The Selenium-Tested Performance Review Portal is a web-based enterprise application designed to digitize the employee appraisal process. The application allows employees to submit self-assessments, managers to evaluate performance, and HR to maintain centralized records. The project additionally demonstrates a complete DevOps pipeline using GitHub, Jenkins, Selenium, Docker, and Ansible.

2. Problem Statement

Many organizations still rely on spreadsheets and manual documentation for employee evaluations. This results in duplicate records, delayed approvals, lack of transparency, and calculation errors. The proposed system provides a secure, centralized, and automated solution.

3. Objectives

- Digitize employee reviews
- Provide secure login
- Enable manager approval workflow
- Automatically calculate performance score
- Maintain centralized HR reports
- Demonstrate CI/CD with automated testing

4. Scope

In Scope: Login, Self Review, Manager Approval, Status Tracking, Performance Score, HR Reports.

Out of Scope: Payroll, Recruitment, Attendance, Leave Management.

5. Stakeholders

Stakeholder	Responsibility
Employee	Submit self review and track status
Manager	Evaluate employees and approve reviews
HR	View reports and analytics
Administrator	Manage users and database

6. Agile Planning

The project follows Scrum methodology with iterative development over five sprints. Each sprint delivers a working increment of the application.

7. Product Backlog

ID	Feature	Priority
PB-01	Employee Login	High
PB-02	Self Review Form	High
PB-03	Manager Dashboard	High
PB-04	Approve / Reject	High
PB-05	Performance Score	Medium
PB-06	HR Reports	Medium
PB-07	Selenium Automation	High
PB-08	Docker Deployment	Medium

8. User Stories

ID	User Story	Acceptance Criteria
US-01	Employee logs into portal	Dashboard opens
US-02	Employee submits review	Review stored
US-03	Manager approves review	Status becomes Approved
US-04	HR views reports	Reports display correctly

9. Sprint Planning

Sprint	Weeks	Deliverable
Sprint 1	1-2	SRS & Planning
Sprint 2	3-4	UI + Database
Sprint 3	5-7	Core Features
Sprint 4	8-10	Selenium + Jenkins
Sprint 5	11-15	Docker + Ansible

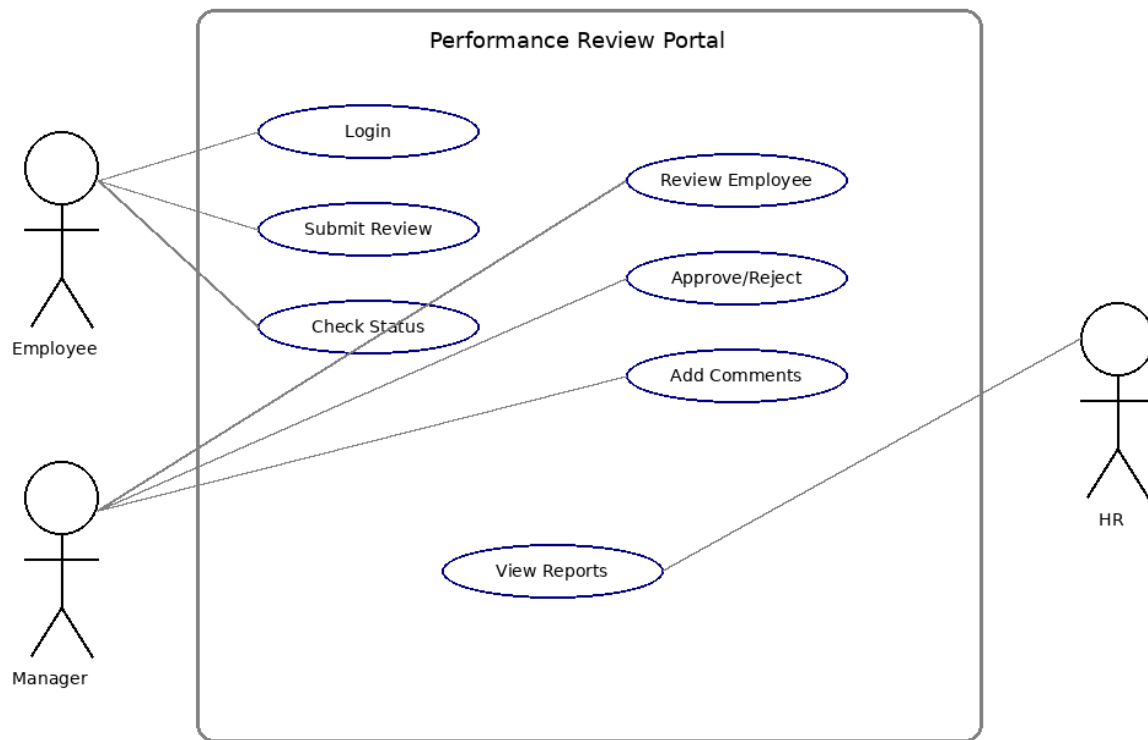
10. Functional Requirements

- User authentication
- Employee profile management
- Self review submission
- Manager evaluation
- Automatic score calculation
- Review history
- HR reporting
- Search employee
- Status tracking

11. Non Functional Requirements

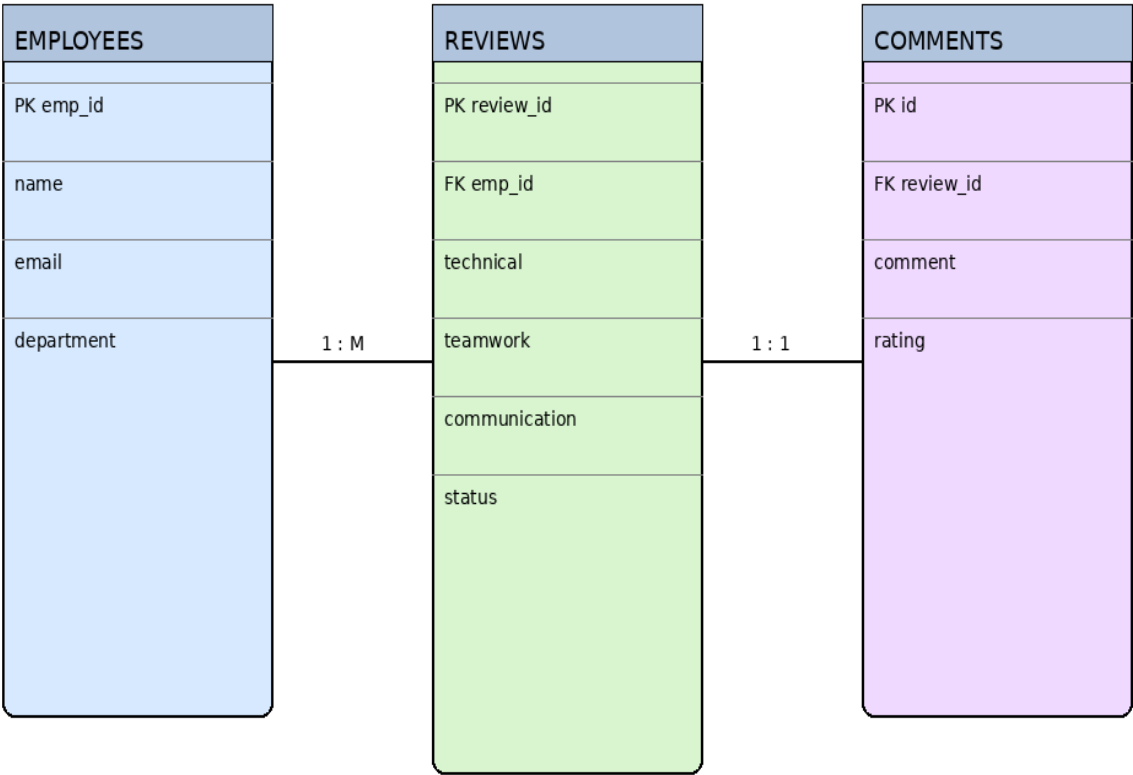
- Secure authentication
- MySQL persistent storage
- Response time below 2 seconds
- Cross browser compatibility
- Scalable architecture
- Automated Selenium testing
- Continuous Integration with Jenkins

12. Use Case Diagram



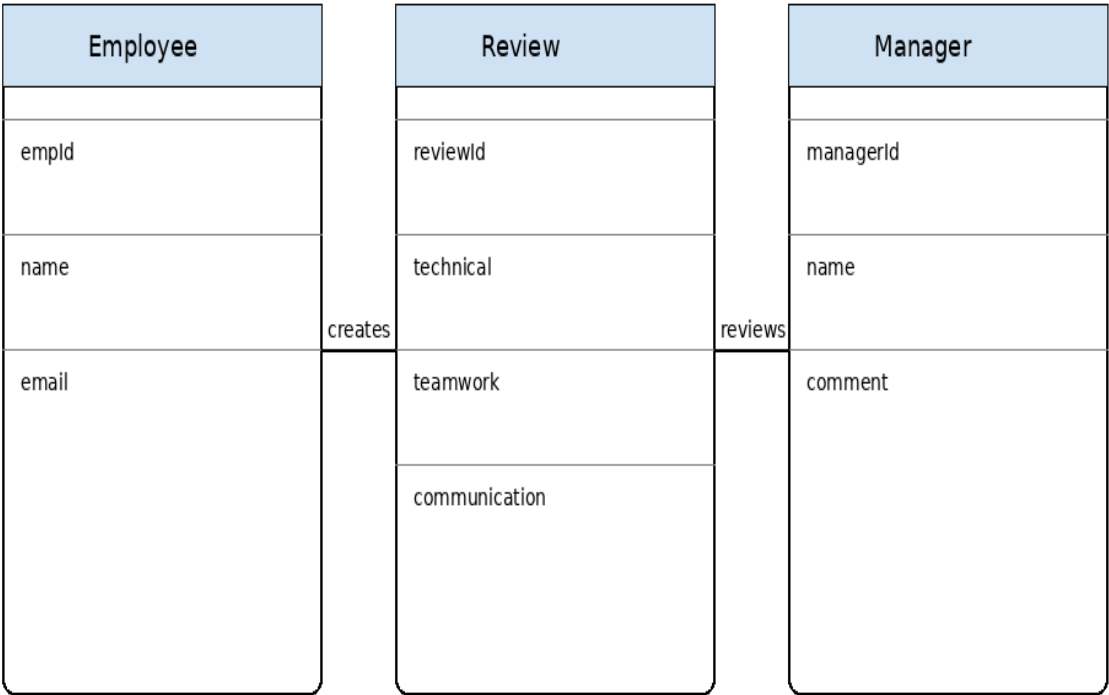
The above diagram represents the system structure and interactions for the Performance Review Portal.

13. ER Diagram



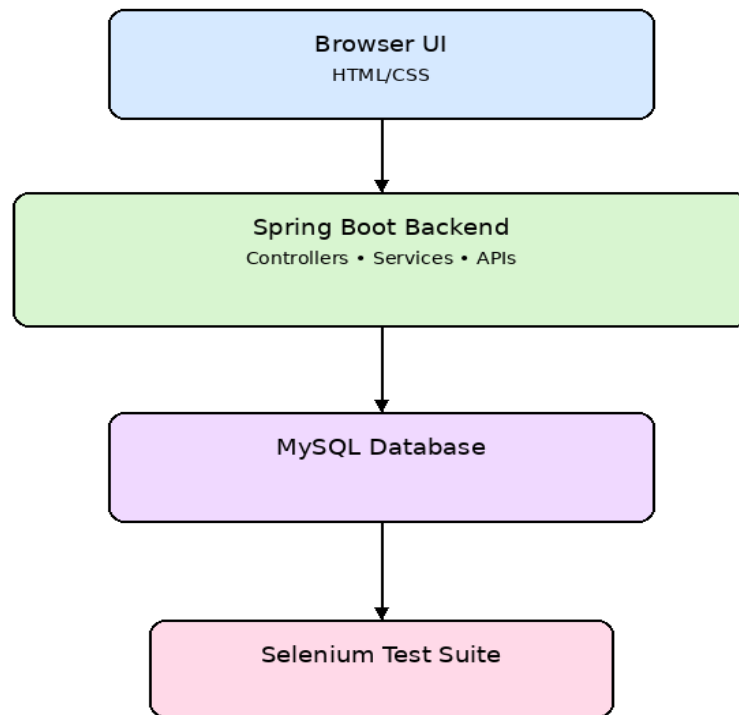
The above diagram represents the system structure and interactions for the Performance Review Portal.

14. Class Diagram



The above diagram represents the system structure and interactions for the Performance Review Portal.

15. System Architecture



The above diagram represents the system structure and interactions for the Performance Review Portal.

16. REST API Design

Method	Endpoint	Description
POST	/login	Authenticate user
GET	/employees	Retrieve employees
POST	/reviews	Submit review
GET	/reviews/{id}	View review
PUT	/reviews/approve	Approve review
GET	/reports	Generate HR reports

17. Database Schema

EMPLOYEE(emp_id PK, name, email, department, designation)

REVIEW(review_id PK, emp_id FK, manager_id FK, communication, technical, teamwork, overall_score, status, review_date)

MANAGER(manager_id PK, name, email, department)

18. Testing Strategy

Selenium automates login, form submission, manager approval, search functionality, and validation testing. Jenkins executes these tests during every build and generates JUnit reports.

19. Risk Analysis

Risk	Mitigation
Database failure	Daily backups
Authentication bugs	Input validation
Build failures	CI testing
Deployment issues	Docker rollback

20. Future Scope

Future enhancements include role-based authorization, email notifications, PDF appraisal reports, analytics dashboards, cloud deployment on AWS, and AI-assisted performance insights.

21. Conclusion

The Selenium-Tested Performance Review Portal provides a complete software engineering and DevOps solution by integrating requirement analysis, database design, web development, automated testing, continuous integration, containerization, and deployment into one enterprise application.